

Router

```
*Aug 20 06:25:12.926: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to do
wn
*Aug 20 06:25:12.931: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to do
wn
*Aug 20 06:25:13.926: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Aug 20 06:25:13.931: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#exit
Router#sh
*Aug 20 06:25:44.151: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip int brief
Interface IP-Address OK? Method Status Prot
ocol
Ethernet0/0 unassigned YES TFTP up up
Ethernet0/1 unassigned YES TFTP up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int e0/0
Router(config-if)#ip add 50.0.0.1 255.0.0.0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#int e0/1
```

Router

```
*Aug 20 06:25:12.926: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to do
wn
*Aug 20 06:25:12.931: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to do
wn
*Aug 20 06:25:13.926: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Aug 20 06:25:13.931: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
```

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#exit
```

```
Router#sh
*Aug 20 06:25:44.151: %SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip int brief
Interface                               IP-Address      OK? Method Status  Prot
ocol
Ethernet0/0                             unassigned      YES TFTP  up      up
Ethernet0/1                             unassigned      YES TFTP  up      up
Ethernet0/2                             unassigned      YES TFTP  down    down
Ethernet0/3                             unassigned      YES TFTP  down    down
```

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int e0/0
Router(config-if)#ip add 50.0.0.1 255.0.0.0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#int e0/1
Router(config-if)#ip add 60.0.0.1 255.0.0.0
Router(config-if)#no sh
Router(config-if)#exit
```

```
Router(config)#exit
```

```
% Invalid input detected at '!' marker.
```

```
Router(config)#exit
```

```
Router#
```

```
*Aug 20 06:30:00.004: %SYS-5-CONFIG_I: Configured from console by console
```

```
Router#
```

```
*Aug 20 06:30:02.565: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
```

```
Router#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from Pfr
```

```
Gateway of last resort is not set
```

```
50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

```
C    50.0.0.0/8 is directly connected, Ethernet0/0
```

```
L    50.0.0.1/32 is directly connected, Ethernet0/0
```

```
60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

```
C    60.0.0.0/8 is directly connected, Ethernet0/1
```

```
L    60.0.0.1/32 is directly connected, Ethernet0/1
```

```
Router#
```



Links



(vwin1)



C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
<C> Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	50.0.0.10
Subnet Mask	:	255.0.0.0
Default Gateway	:	:

C:\Documents and Settings\user>_

QEMU (Win2)



My Documents



My Computer



My Network Places



Recycle Bin



Internet Explorer

on C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	60.0.0.10
Subnet Mask	:	255.0.0.0
Default Gateway	:	:

C:\Documents and Settings\user>_

C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:	50.0.0.10
IP Address	:	:	255.0.0.0
Subnet Mask	:	:	:
Default Gateway	:	:	:

C:\Documents and Settings\user>ping 50.0.0.1

Pinging 50.0.0.1 with 32 bytes of data:

Reply from 50.0.0.1:	bytes=32	time=1ms	TTL=255
Reply from 50.0.0.1:	bytes=32	time=1ms	TTL=255
Reply from 50.0.0.1:	bytes=32	time<1ms	TTL=255
Reply from 50.0.0.1:	bytes=32	time<1ms	TTL=255

Ping statistics for 50.0.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>


```
C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . .               : 60.0.0.10
    Subnet Mask . . . . .             : 255.0.0.0
    Default Gateway . . . . .         : 

C:\Documents and Settings\user>ping 60.0.0.1

Pinging 60.0.0.1 with 32 bytes of data:

Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>
```



My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco OM-I...

va ASDM

TTY

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>
C:\Documents and Settings\user>cd\

C:\>ping 60.0.0.1

Pinging 60.0.0.1 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```


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QEMU (Win2)

My Documents

My Computer

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Recycle Bin

Internet Explorer

C:\WINDOWS\system32\cmd.exe

```
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

C:\Documents and Settings\user>cd\

C:\>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

```
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
```

```
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

C:\>

QEMU (Win1)

C:\WINDOWS\system32\cmd.exe

Ping statistics for 60.0.0.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss).

C:\>route print

Interface List

0x1 MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	50.0.0.0	255.0.0.0	50.0.0.10	50.0.0.10	10
	50.0.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10
50.255.255.255	255.255.255.255		50.0.0.10	50.0.0.10	10
	127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
	224.0.0.0	240.0.0.0	50.0.0.10	50.0.0.10	10
255.255.255.255	255.255.255.255		50.0.0.10	50.0.0.10	1

Persistent Routes:

None

C:\>

QEMU (Win2)

C:\WINDOWS\system32\cmd.exe

Ping statistics for 50.0.0.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>route print

Interface List

0x1 MS TCP Loopback interface
0x2 ...50 86 5f 09 50 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
60.0.0.0		255.0.0.0	60.0.0.10	60.0.0.10	10
60.0.0.10		255.255.255.255	127.0.0.1	127.0.0.1	10
60.255.255.255		255.255.255.255	60.0.0.10	60.0.0.10	10
127.0.0.0		255.0.0.0	127.0.0.1	127.0.0.1	1
224.0.0.0		240.0.0.0	60.0.0.10	60.0.0.10	10
255.255.255.255		255.255.255.255	60.0.0.10	60.0.0.10	1

Persistent Routes:

None

C:\>

EMU (Win1)

```
C:\WINDOWS\system32\cmd.exe

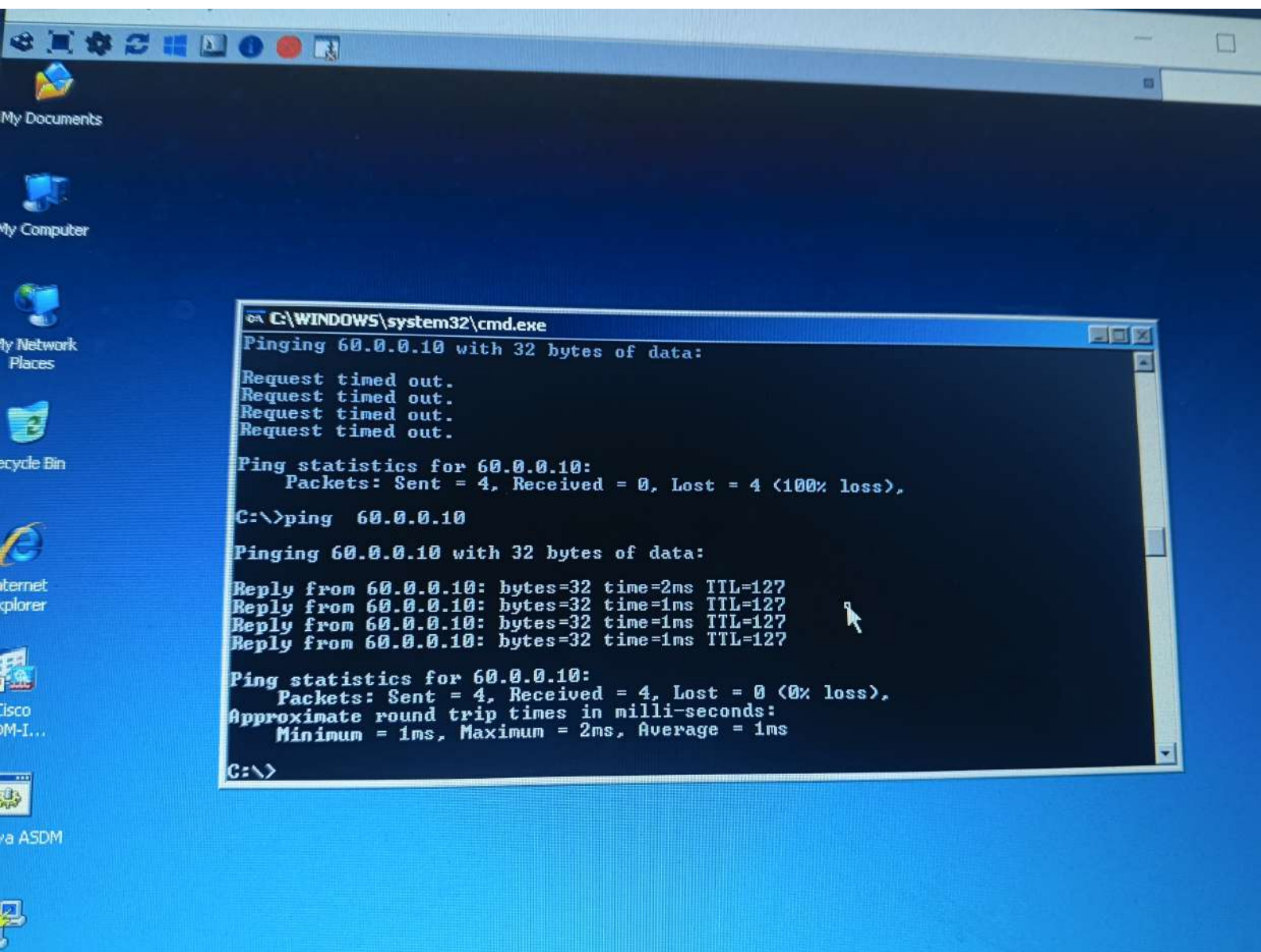
C:\>route add 60.0.0.0 mask 255.0.0.0 50.0.0.1

C:\>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====

Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
50.0.0.0                255.0.0.0        50.0.0.10       50.0.0.10        10
50.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1        10
50.255.255.255         255.255.255.255  50.0.0.10       50.0.0.10        10
60.0.0.0                255.0.0.0        50.0.0.1        50.0.0.10        1
127.0.0.0               255.0.0.0        127.0.0.1       127.0.0.1        1
224.0.0.0               240.0.0.0        50.0.0.10       50.0.0.10        10
255.255.255.255        255.255.255.255  50.0.0.10       50.0.0.10        1
=====

Persistent Routes:
None

C:\>
```



C:\WINDOWS\system32\cmd.exe

Pinging 60.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 60.0.0.10:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 60.0.0.10

Pinging 60.0.0.10 with 32 bytes of data:

Reply from 60.0.0.10: bytes=32 time=2ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 60.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>

QEMU (Win2)

C:\WINDOWS\system32\cmd.exe

Persistent Routes:
None

C:\>add route 50.0.0.0 mask 255.0.0.0 60.0.0.10
'add' is not recognized as an internal or external command,
operable program or batch file.

C:\>route add 50.0.0.0 mask 255.0.0.0 60.0.0.1

C:\>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

Reply from 50.0.0.10: bytes=32 time<1ms TTL=127

Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

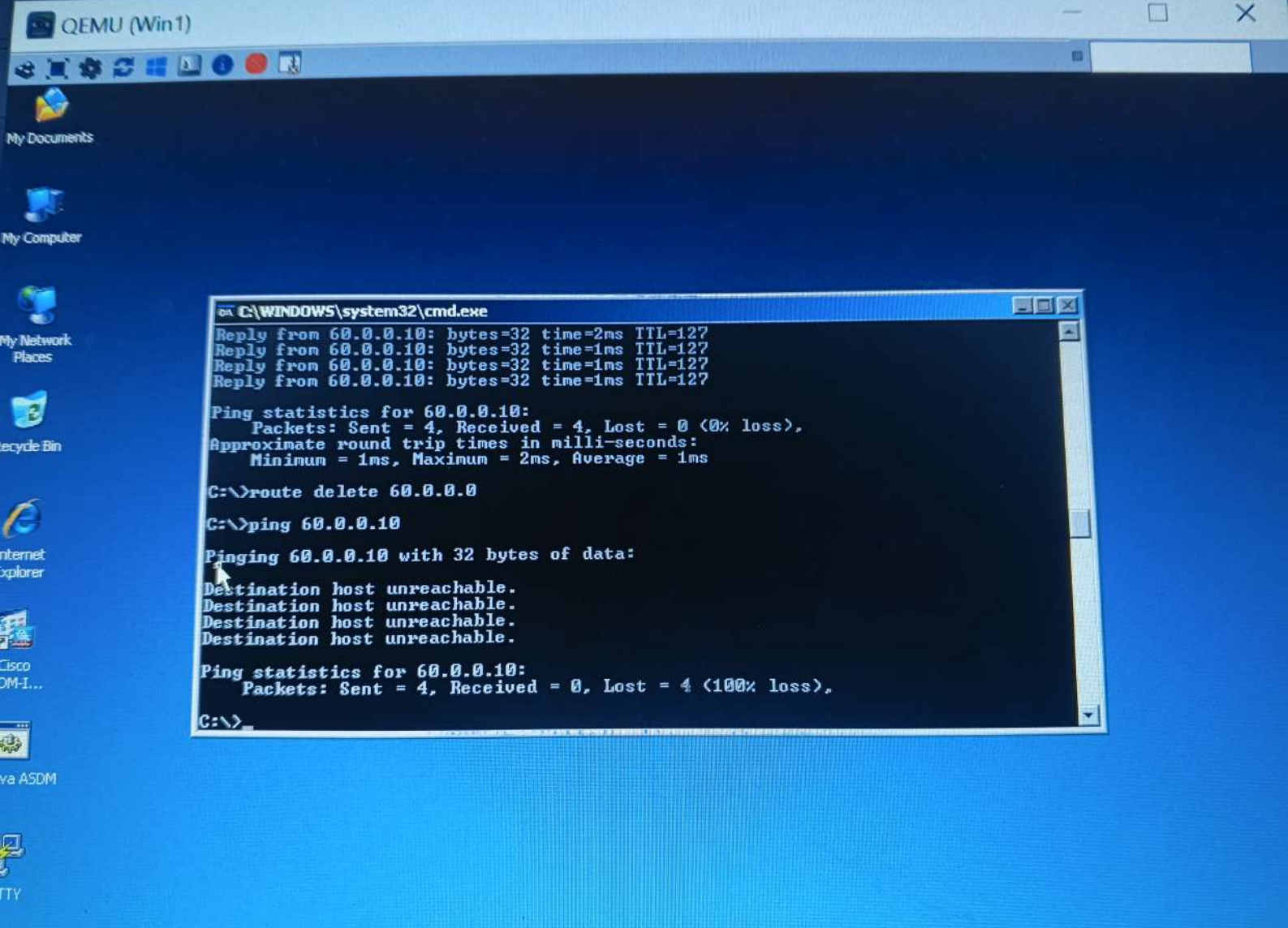
Ping statistics for 50.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>



unassigned
unassigned
commands, one per
0/0
add 50.0.0.1 255
sh
t
/1
add 60.0.0.1 255
sh
c
ed at '^' marke
SYS-5-CONFIG_I
PNP-6-PNP_DISC
connected, S -
EIGRP external

QEMU (Win2)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

C:\WINDOWS\system32\cmd.exe

```
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time<1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>route delete 50.0.0.0

C:\>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

QEMU (Win1)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASDM

C:\WINDOWS\system32\cmd.exe

C:\>ncpa.cpl

C:\>route print

Interface List

0x1 MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	50.0.0.1	50.0.0.10	10
	50.0.0.0	255.0.0.0	50.0.0.10	50.0.0.10	10
	50.0.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10
50.255.255.255	255.255.255.255	255.255.255.255	50.0.0.10	50.0.0.10	10
	127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
	224.0.0.0	240.0.0.0	50.0.0.10	50.0.0.10	10
255.255.255.255	255.255.255.255	255.255.255.255	50.0.0.10	50.0.0.10	1

Default Gateway: 50.0.0.1

Persistent Routes:

None

C:\>

QEMU (Win2)

C:\WINDOWS\system32\cmd.exe

C:\>ncpa.cpl

C:\>route print

Interface List

0x1 MS TCP Loopback interface
0x2 ...50 86 5f 09 50 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	60.0.0.1	60.0.0.10	10
	60.0.0.0	255.0.0.0	60.0.0.10	60.0.0.10	10
	60.0.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10
	60.255.255.255	255.255.255.255	60.0.0.10	60.0.0.10	10
	127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
	224.0.0.0	240.0.0.0	60.0.0.10	60.0.0.10	10
	255.255.255.255	255.255.255.255	60.0.0.10	60.0.0.10	1

Default Gateway: 60.0.0.1

Persistent Routes:

None

C:\>

MU (Win1)

```
C:\WINDOWS\system32\cmd.exe
 50.0.0.10 255.255.255.255 127.0.0.1 127.0.0.1 10
50.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 10
127.0.0.0 255.0.0.0 127.0.0.1 127.0.0.1 1
224.0.0.0 240.0.0.0 50.0.0.10 50.0.0.10 10
255.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 1
Default Gateway: 50.0.0.1
=====
Persistent Routes:
None
C:\>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Reply from 60.0.0.10: bytes=32 time=3ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\>
```


QEMU (Win2)

C:\WINDOWS\system32\cmd.exe

60.0.0.10	255.255.255.255	127.0.0.1	127.0.0.1	10
60.255.255.255	255.255.255.255	60.0.0.10	60.0.0.10	10
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
224.0.0.0	240.0.0.0	60.0.0.10	60.0.0.10	10
255.255.255.255	255.255.255.255	60.0.0.10	60.0.0.10	1

Default Gateway: 60.0.0.1

Persistent Routes:
None

C:\>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time=2ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 50.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>

Internet
Explorer



Cisco
ASDM-I...



Old Java ASDM